Disorders of the Spine & Spinal Cord

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Outlines

Intrinsic diseases of the spinal cord

Congenital

Infective/Inflammatory

Vascular

Neoplastic

Metabolic

Degenerative

Spinal cord compression

Objectives

- To understand the **anatomical stratification** of spinal cord lesions.
- To learn the **clinical features** of intrinsic & some of the extrinsic cord diseases.
- To apply anatomy for the **localization** of the level of cord compression.
- To be able to evaluate the **differential diagnosis** of these disordres.
- To learn how to investigated for cord & spine disease.
- To be able **to treat cord compression urgently** before permanent damage occur.
- To find the **treatable causes** & apply accordingly.

Disorders of the Spine & Spinal Cord

- The spinal cord & spinal roots may be affected by **intrinsic** disease

  or by disorders of the surrounding **meninges** & **bones**.

- The clinical presentation depends on the **anatomical** level, as well as the nature of the **pathological** process involved.

- It is important to recognize that spinal cord **compression** need **urgent** action.

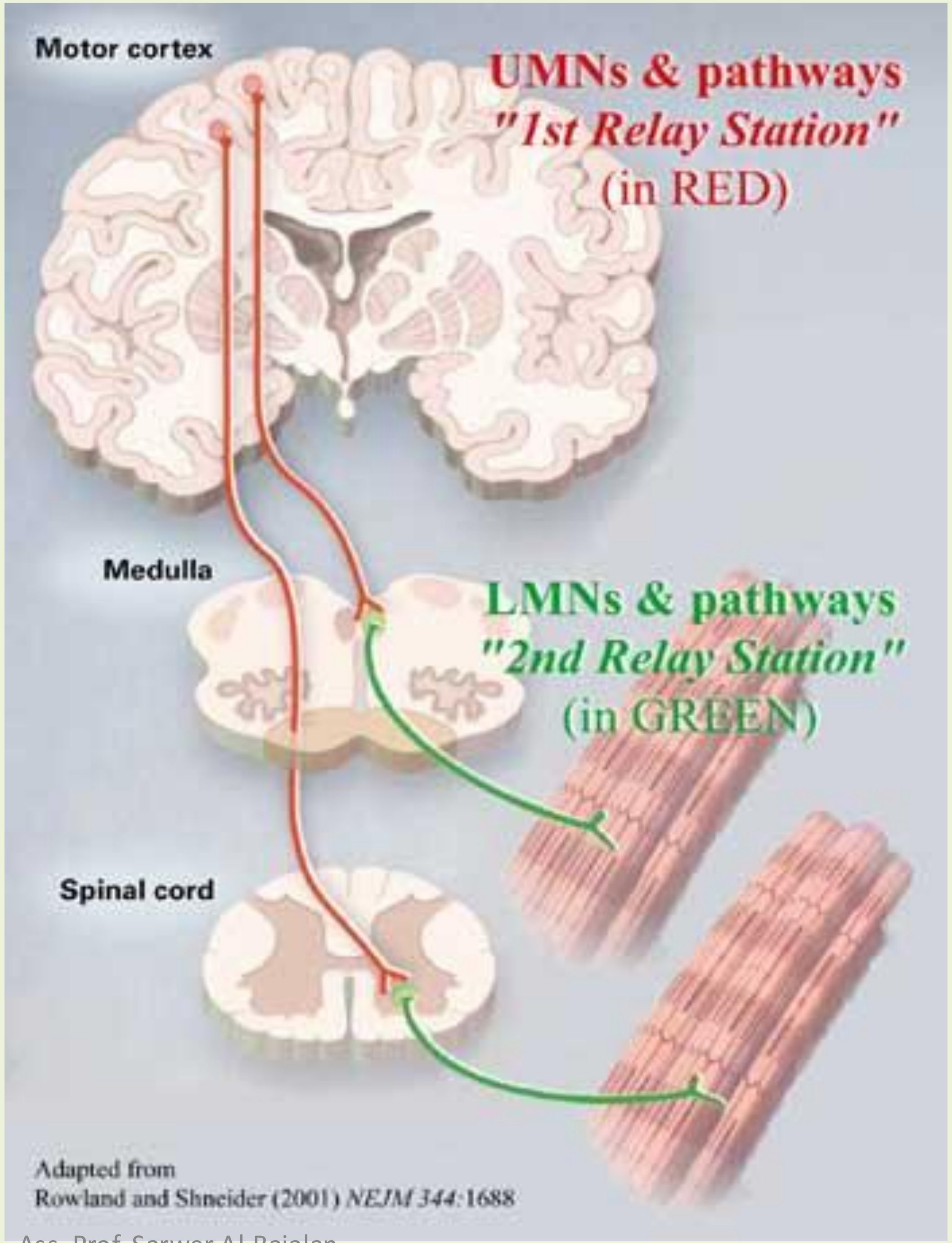
Upper vs. Lower Motor Neuron Lesion

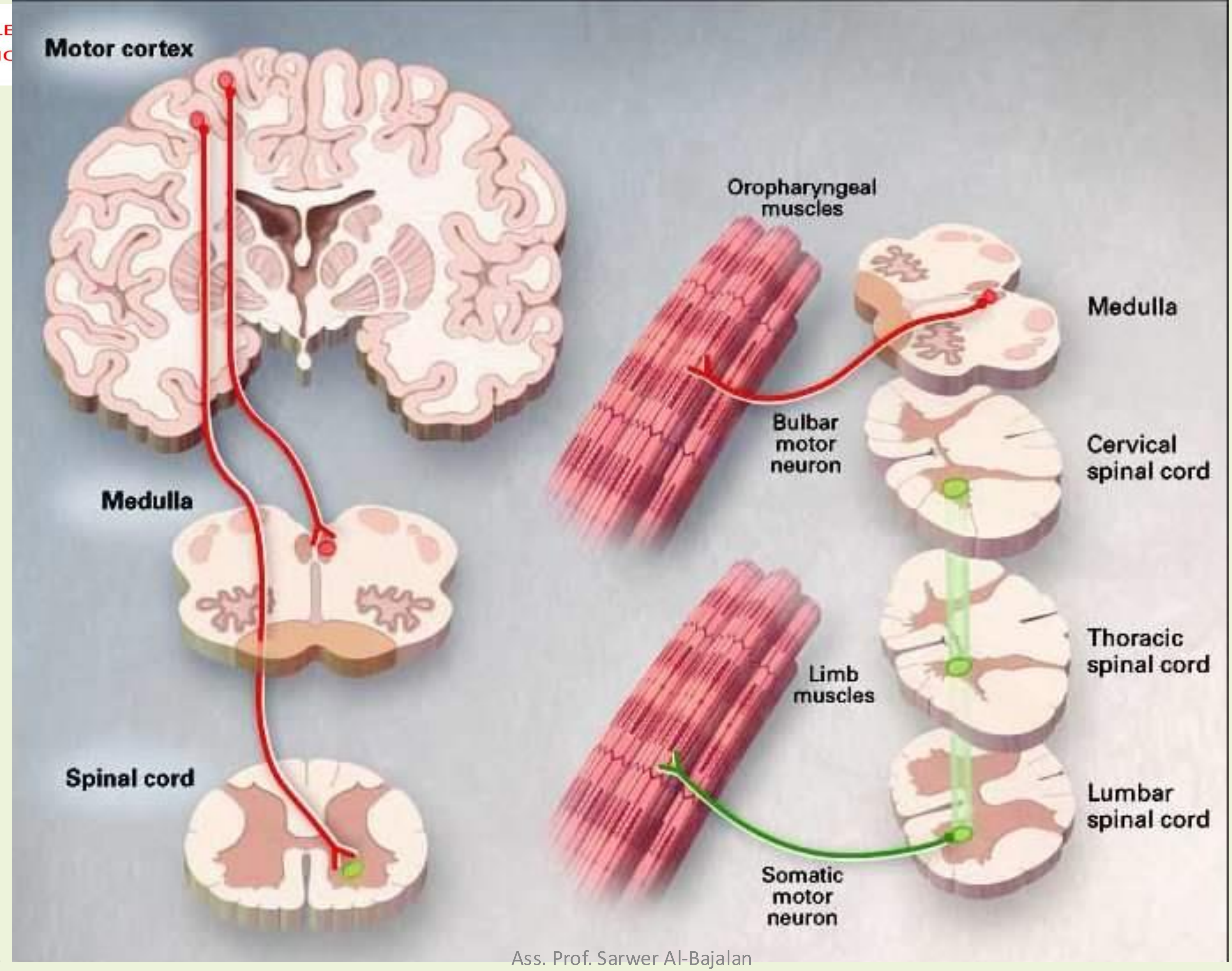
- **Upper motor neuron lesion**
  - Motor cortex → internal capsule → brainstem → spinal cord

- **Lower motor neuron lesion**
  - Anterior horn cell → n. root → plexus → peripheral n. → neuromuscular junction → muscle

Pyramidal (Corticospinal) Tract (UMN): >>>

LMN Fibers: >>>

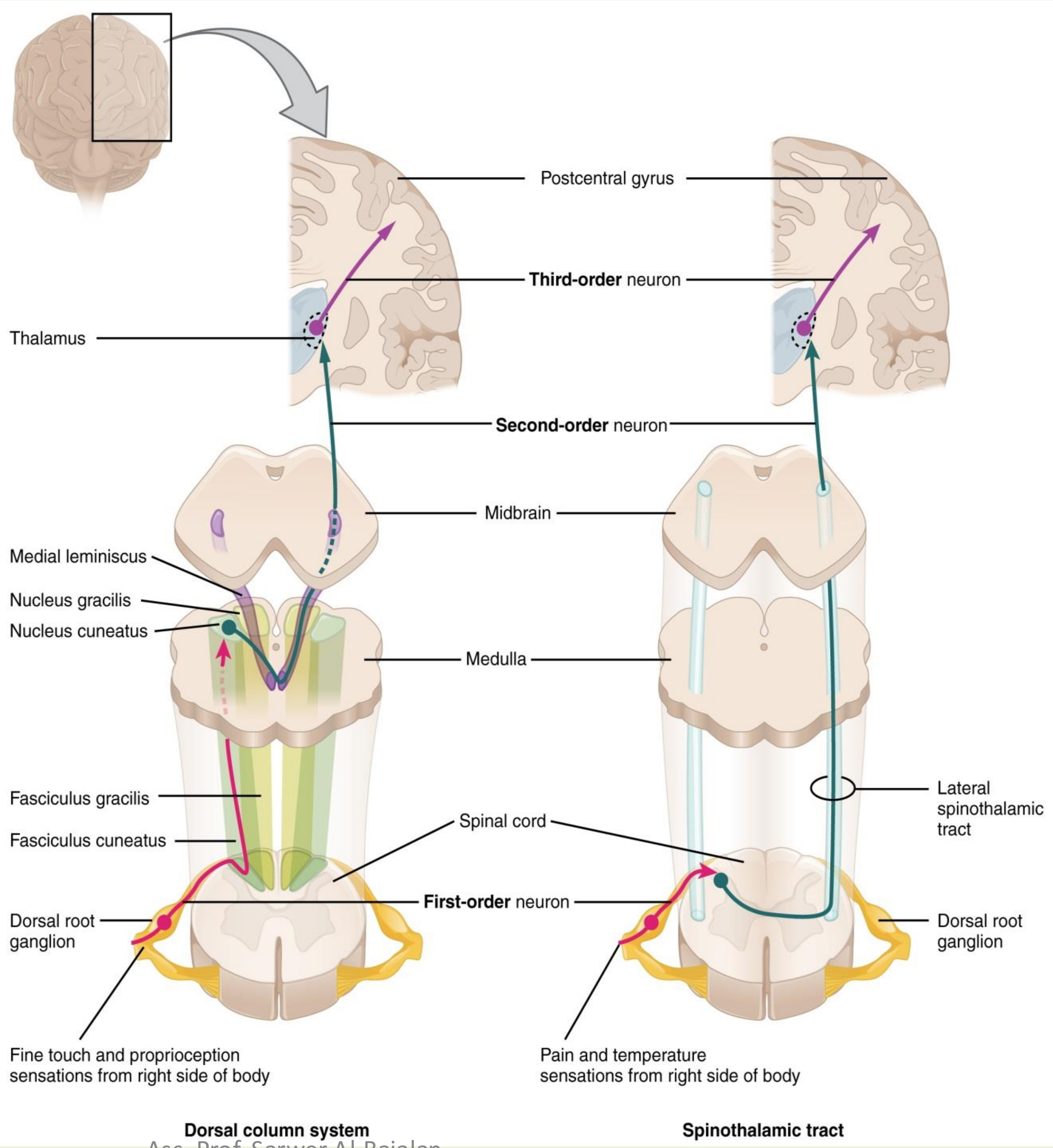




Motor cortex
Medulla
Spinal cord
Oropharyngeal muscles
Medulla
Bulbar motor neuron
Cervical spinal cord
Thoracic spinal cord
Limb muscles
Lumbar spinal cord
Somatic motor neuron

Dorsal column:
(Vibration & Joint Position)

Spinothalamic Tract:
(Pain & Temperature)



Basic Features of Spinal Cord Disease

- UMN findings below the lesion
  - Pyramidal weakness
  - Hyperreflexia and Babinski's sign
  - Spasticity
  - Disuse atrophy(late)

- Sensory and motor involvement that localizes to a spinal cord level

- Bowel and Bladder dysfunction common

- Remember that the spinal cord ends at about T12-L1

History

- Onset
  - Acute, subacute, chronic

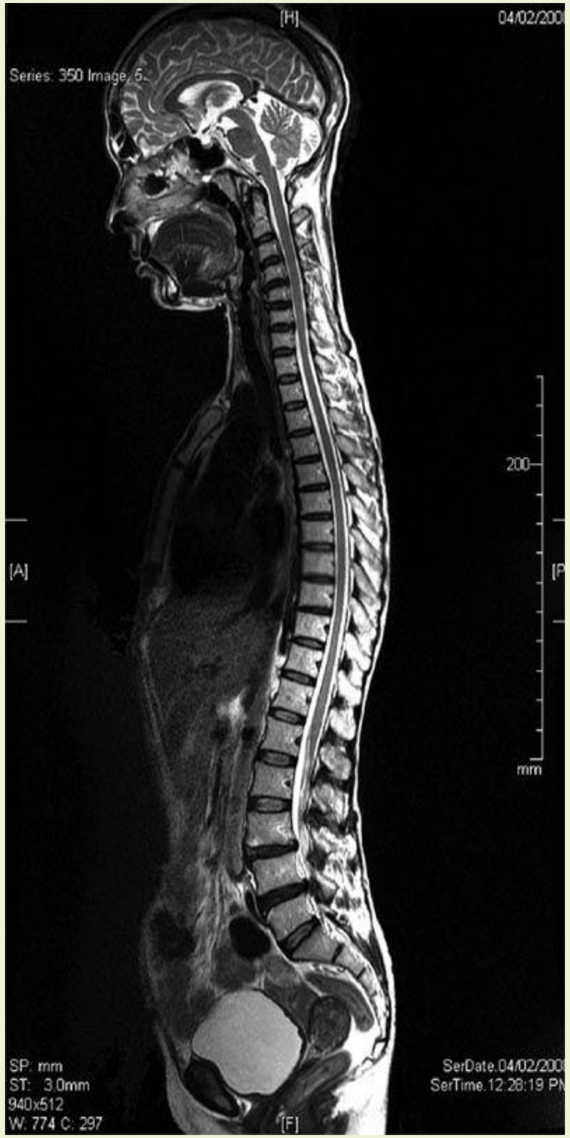
- Symptoms
  - Pain
  - Weakness
  - Sensory
  - Autonomic
- Past history
- Family history

Tempo of Spinal Cord Disease

	Acute	Subacute	Chronic
Trauma	X		
Mass lesion	X	X	
Infectious	X	X	X
Inherited			X
Vascular	X	X	X
Autoimmune	X	X	
Nutritional			X

Motor Exam

- Strength - helps to localize the lesion
  - Upper cervical (Above C5)
    - Quadriplegia with impaired respiration
  - Lower cervical
    - Proximal arm strength preserved
    - Hand weakness and leg weakness
  - Thoracic
    - Paraplegia
    - (**DDx**: we can also see paraplegia
  with a midline lesion in the brain)
- Tone
  - Increased distal to the lesion



Sensory Exam

- Establish a sensory level
  - Dermatomes
    - C2 angle of the jaw
    - C5- T2 (see upper limbs dermatomes)
    - Nipples: T4
    - Xiphi sternum: T6
    - Umbilicus: T10
    - Inguinal ligament: L1
- Posterior columns
  - Vibration
  - Joint position sense (proprioception)
- Spinothalamic tracts
  - Pain
  - Temperature

Autonomic disturbances

- Neurogenic bladder
  - Urgency, incontinence, retention
- Bowel dysfunction
  - Constipation more frequent than incontinence
- With a high cord lesion, loss of blood pressure control
- Alteration in sweating

Investigation of Spinal Cord Disease

- Radiographic exams
  - Plain films
  - Myelography
  - CT scan with myelography
  - MRI
- Spinal tap
  - If you suspect: inflammation, multiple sclerosis, rupture of a vascular malformation

Etiology of Spinal Cord Disease

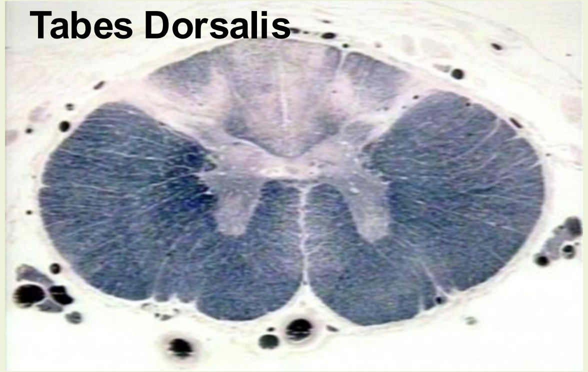
B12 Deficiency
(Subacute Combined Degeneration of the Cord)

- ❑ **Causes:**
  - malabsorption of B12 secondary to pernicious anemia or surgery
  - insufficient dietary intake
- ❑ **Features:**
  - Posterior columns (JP & Vibration loss)
  - CST involvement(Extensor Plantars)
  - with a superimposed peripheral neuropathy (absent ankles) *

Infections Involving the Spinal Cord

- Poliomyelitis
  - only the anterior horn cells are infected

- Tabes dorsalis
  - dorsal root ganglia and dorsal columns are involved
  - tertiary syphilis
  - sensory ataxia, “lightening pains”
- HIV myelopathy
  - mimics B12 deficiency
- HTLV-1 myelopathy -
  - tropical spastic paraparesis



Demyelinating Diseases

Multiple Sclerosis:

- Demyelination is the underlying pathology
- Cord disease can be presenting feature of MS or occur at any time during the course of the disease
- Lesion can be at any level of the cord
  - Patchy
  - Transverse

Devic's Disease: or neuromyelitis optica**(NMO)**

- Transverse myelitis with optic neuritis

Vascular Diseases of the Spinal Cord

Anterior spinal artery infarct

- **Causes**:
  - Atherosclerosis
  - Iatrogenic: during surgery in which the aorta is clamped
  - Dissecting aortic aneurysm
  - Chronic meningitis
  - Trauma

- **Features**:
  - Weakness (CST) and pain/temperature loss (spinothalamic tracts)
  - Posterior columns preserved (JPS, & vibration are not affected)

Vascular Diseases of the Spinal Cord, *cont*

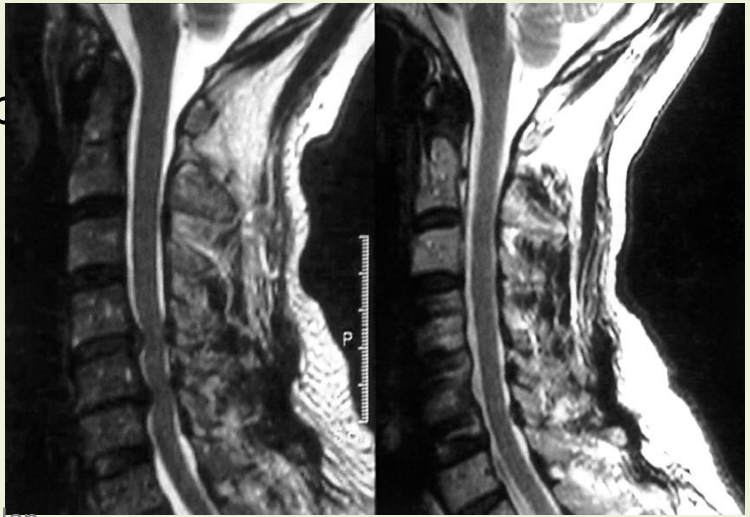
Arteriovenous malformation (AVM) & venous angiomas:

- Both occur in primarily the thoracic cord
- May present either acutely, subacutely or chronically (act as a compressive lesion)
- Can cause recurrent symptoms (MS like)
- If they bleed
  - Associated with pain and bloody CSF
- Notoriously difficult to diagnose

- **Hematoma:** trauma, occasionally tumor

Other Disease of the Spinal Cord

- Hereditary spastic paraparesis
  - Usually autosomal dominant
- Infectious process of the vertebrae
  - TB, bacterial
- Herniated disc with cord compression
  - Most herniated discs are lateral and only compress a nerve root
- Degenerative disease of the vertebrae
  - Cervical spondylosis with a myelop
  - Spinal stenosis



Classical spinal cord syndromes

- Anterior spinal artery infarct
- Brown Sequard syndrome
- Syringomyelia
- Conus medullaris/caude equina lesions

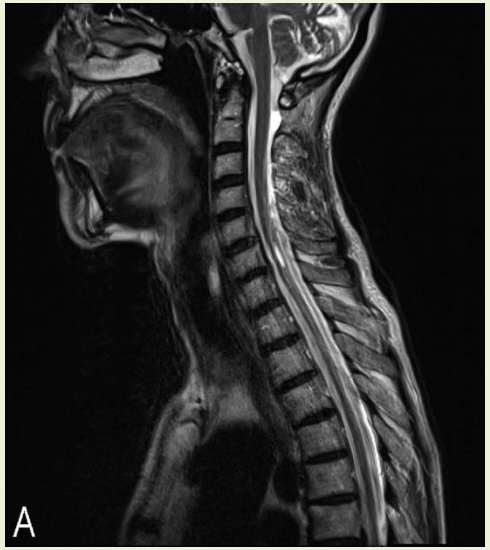
Brown Sequard Syndrome (Cord hemisection):

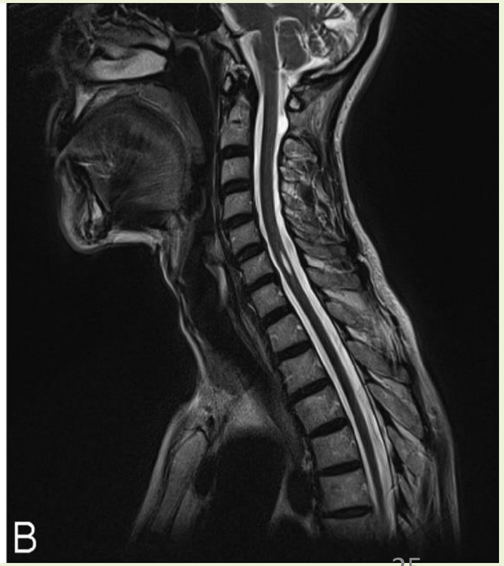
- **Causes:** Trauma, tumor or vascular
- **C/F:**
  - **Dissociated sensory loss**
    - Loss of pain and temperature **contralateral** to lesion, one or 2 levels below
      - crossing of spinothalamic tracts 1-2 segments above where they enter
    - Loss of vibration/proprioception **ipsilateral** to the lesion
      - these pathways cross at the upper medulla
  - Weakness and **UMN** findings **ipsilateral** to lesion

Syringomyelia

Fluid filled cavitation in the center of the cord

- **Causes:** trauma or in association with Arnold Chiari malformation
- **Site:** Cervical cord most common
- **C/F:**
  - Loss of pain and temperature related to the crossing fibers occurs early (cape like sensory loss)
  - Weakness of muscles in arms with atrophy and hyporeflexia (AHC)
  - CST involvement with brisk reflexes in the legs, spasticity, and weakness (Late)





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Conus Medullaris vs. Cauda Equina Lesion

Finding	Conus medullaris	Cuda Equina
Motor	Symmetric	Asymmetric
Sensory loss	Saddle	Saddle
Pain	Uncommon	Common
Reflexes	Increased	Decreased
Bowel/bladder	Common	Uncommon

BOF

All of the following are true regarding causes of spinal cord disorders except:

A. Inherited disorders usually cause chronic dysfunction.

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B. Brown Sequard Syndrome means loss of all sensory modalities below the lesion site.

C. B12 Deficiency may cause absent ankle jerks with Babinski sign.

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Examples of Intrinsic diseases of the spinal cord

- Congenital
- Infective/Inflammatory
- Vascular
- Neoplastic
- Metabolic
- Degenerative

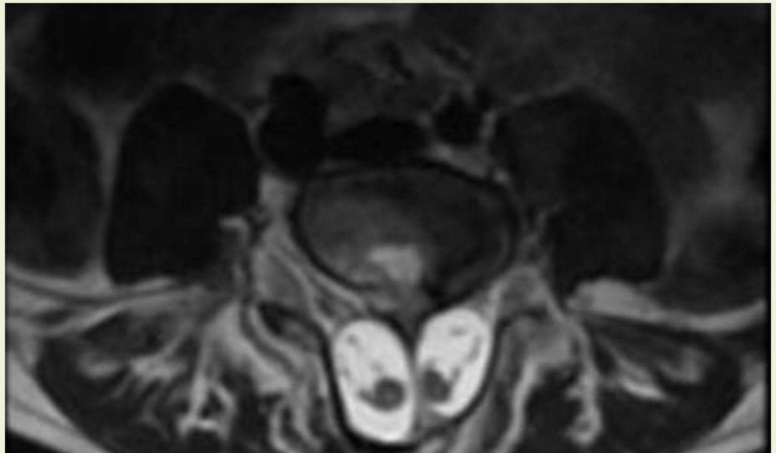
Type of disorder	Condition	Clinical features
Congenital	Diastematomyelia (spina bifida)	Features variably present at birth and deteriorate thereafter LMN features, deformity and sensory loss of legs Impaired sphincter function Hairy patch or pit over low back Incidence reduced by increased maternal intake of folic acid during pregnancy
	Hereditary spastic paraplegia	Onset usually in adult life Autosomal dominant inheritance usual Slowly progressive UMN features affecting legs > arms Little or no sensory loss
Infective/inflammatory	Transverse myelitis due to viruses (HZV), schistosomiasis, HIV, MS, sarcoidosis	Weakness and sensory loss, often with pain, developing over hours to days UMN features below lesion Impaired sphincter function
	Paraneoplastic	May predate tumour diagnosis
Vascular	Anterior spinal artery infarct due to atherosclerosis, aortic dissection, embolus	Abrupt onset Anterior horn cell loss (LMN) at level of lesion UMN features below it Spinothalamic sensory loss below lesion but dorsal column sensation spared
	Spinal AVM/dural fistula	Onset variable (acute to slowly progressive) Variable LMN, UMN, sensory and sphincter disturbance Symptoms and signs often not well localised to site of AVM
Neoplastic	Glioma, ependymoma	Weakness and sensory loss often with pain, developing over months to years UMN features below lesion in cord; additional LMN features in conus Impaired sphincter function
Metabolic	Vitamin B_{12} deficiency (subacute combined degeneration)	Progressive spastic paraparesis with proprioception loss Absent reflexes due to peripheral neuropathy ± Optic nerve and cerebral involvement
	Copper deficiency	Excess dietary zinc
	Nitrous oxide toxicity	Modifies vitamin B_{12} metabolism
Degenerative	Motor neuron disease	Relentlessly progressive LMN and UMN features, associated bulbar weakness No sensory involvement
	Syringomyelia	Gradual onset over months or years, pain in cervical segments Anterior horn cell loss (LMN) at level of lesion, UMN features below it Suspended spinothalamic sensory loss at level of lesion, dorsal columns preserved (see Figs. 28.18F and 28.49)

(AVM = arteriovenous malformation; HIV = human immunodeficiency virus; HZV = herpes zoster virus; LMN = lower motor neuron; MS = multiple sclerosis; UMN = upper motor neuron)

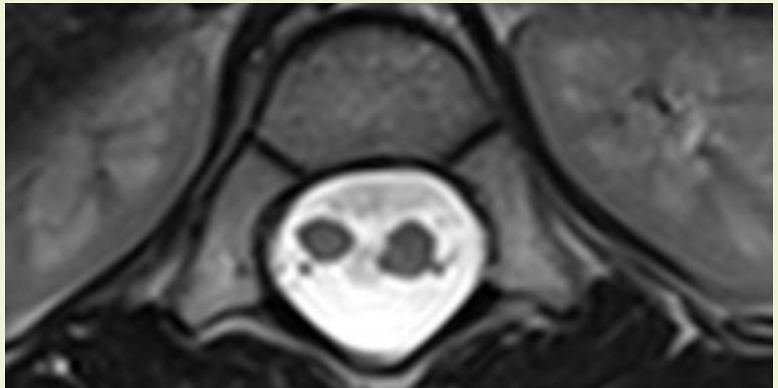
Congenital - Diastematomyelia (spina bifida)

- **Diastematomyelia** (**split cord malformation**) refers to a type of **Spinal Dysraphism** (spina bifida occulta) characterized by a longitudinal split in the spinal cord.

- **Features** variably present at birth & deteriorate Lator.
- **LMN** features, deformity & sensory loss of legs.
- Impaired **sphincter** function.
- Hairy patch or pit over low back.
- Incidence reduced by maternal intake of **folic acid**.



Diastematomyelia Type I (Double Dural Sac)



Diastematomyelia Type II (Single Dural Sac)

Hereditary spastic paraplegia

- Onset usually in adult life
- Autosomal dominant inheritance usual
- Slowly progressive UMN features affecting legs > arms
- Little or no sensory loss

Spinal AVM/dural Fistula

- Onset variable (acute to slowly progressive)
- Variable LMN, UMN, sensory & sphincter disturbance
- Symptoms & signs often not well localized to site of AVM

Metabolic

Examples:

- Vitamin B 12 deficiency (subacute combined degeneration)
- Copper deficiency (Excess dietary zinc intake)
- Nitrous oxide toxicity (Modifies vitamin B 12 metabolism)

Features:

- Progressive spastic paraparesis with proprioception loss
- Absent reflexes due to peripheral neuropathy
- ±Optic nerve & cerebral involvement

Spinal cord compression - 1

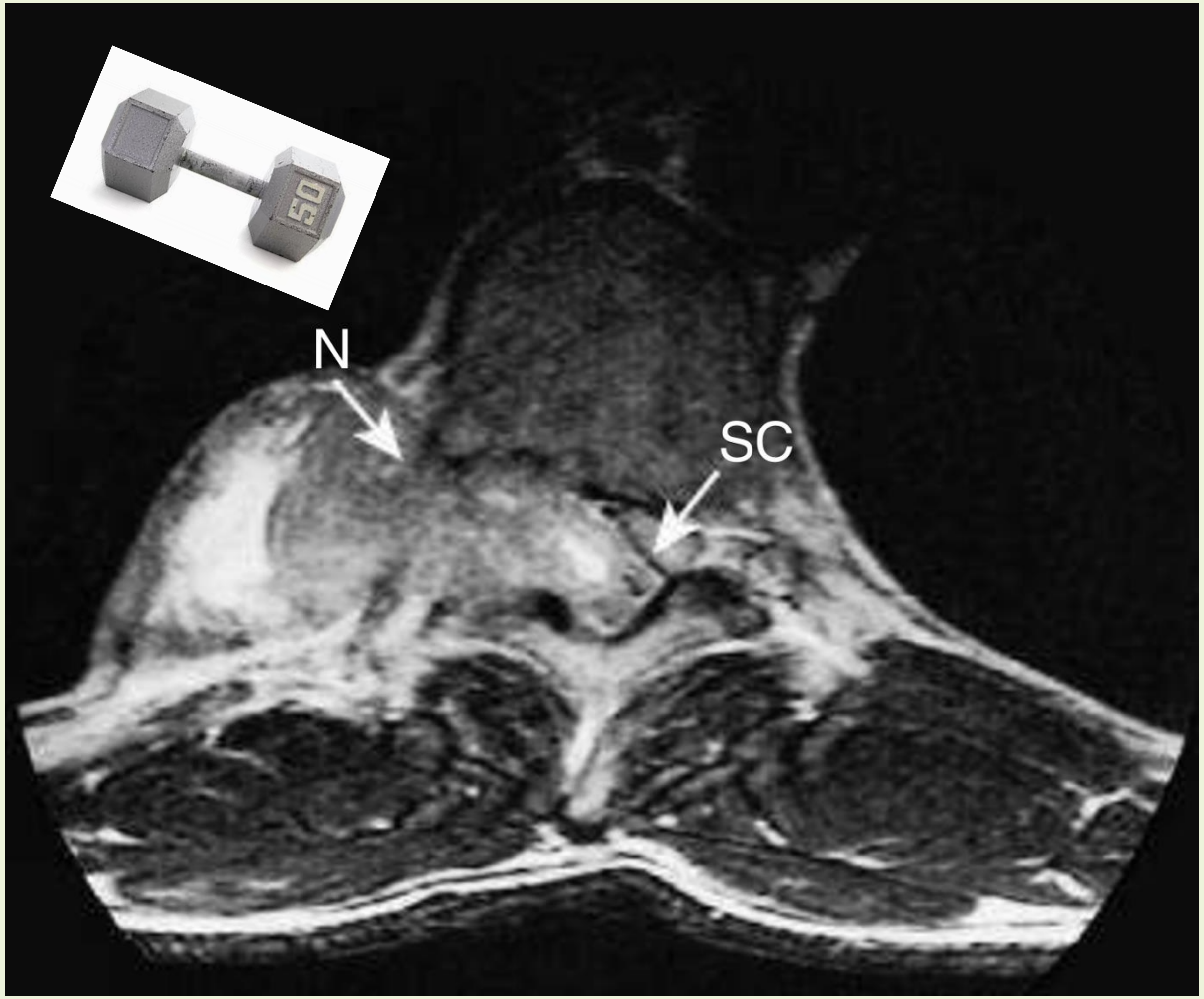
- Spinal cord compression is one of the common neurological emergencies.
- A mass in spinal canal may damage nerves either **directly** or **indirectly** through blood supply.
- **Edema** from venous obstruction **impairs** neuronal function, & arterial **ischemia** may lead to cord **necrosis** .
- Early damage are reversible but severely damaged neurons do not recover.

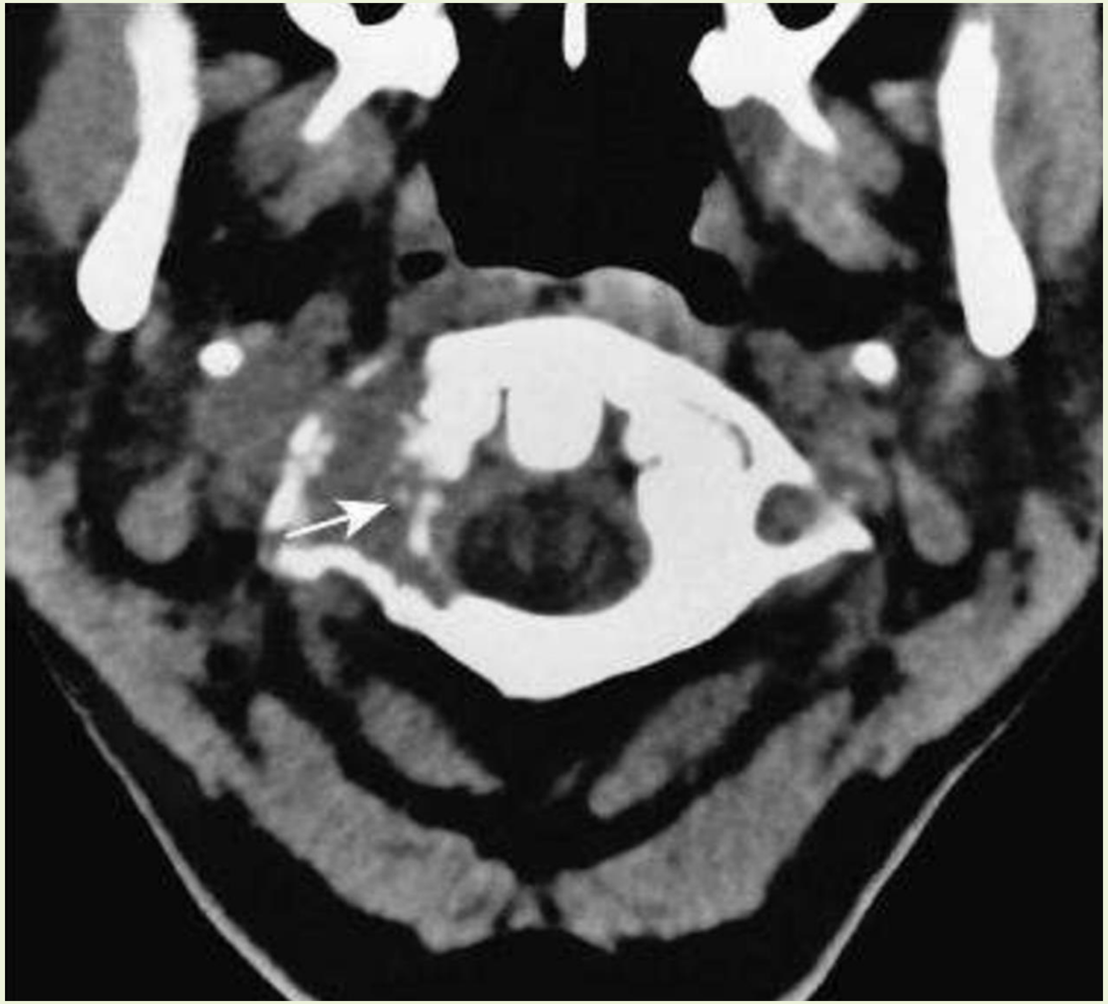
Causes of spinal cord compression:

Site	Frequency	Causes
Vertebral:	**80%**	**Trauma (extradural)** **Intervertebral disc prolapse** **Metastatic carcinoma** (e.g. breast, prostate, bronchus) **Myeloma** **Tuberculosis**
Meninges: (intradural, extramedullary)	**15%**	**Tumours** (e.g. meningioma, neurofibroma, ependymoma, metastasis, lymphoma, leukaemia) **Epidural abscess**
Spinal cord: (intradural, intramedullary)	**5%**	**Tumours** (e.g. glioma, ependymoma, metastasis)

Spinal cord compression - Clinical features

- The onset of symptoms of spinal cord compression is usually slow (over weeks) but can be acute as a result of trauma or metastases, especially if there is associated arterial occlusion.
- Pain & sensory symptoms occur early, while weakness & sphincter dysfunction are usually late manifestations.
- The signs vary according to the level of the cord compression & the structures involved.
- There may be tenderness to percussion over the spine if there is vertebral dis ease & this may be associated with a local kyphosis.
- Involvement of the roots at the level of the compression may cause dermatomal sensory impairment & corresponding lower motor signs.
- Interruption of fibres in the spinal cord causes sensory loss & upper motor neuron signs below the level of the lesion, & there is often disturbance of sphincter function.
- The Brown–Séquard syndrome (see Fig. 28.18E) results if damage is confined to one side of the cord; the findings are explained by the anatomy of the sensory tracts (see Fig. 28.11).
- With compressive lesions, there is usually a band of pain at the level of the lesion in the distribution of the nerve roots subject to compression.





Axial MRI of thoracic spine. A neurofibroma (N) is compressing the spinal cord (SC) & emerging in a ‘dumbbell’ fashion through the vertebral foramen into the paraspinal space.

CT myelogram of cervical spine at the level of C2 showing bony erosion of vertebra by a metastasis (arrow).

Spinal cord compression - Investigations

- Acute or subacute spinal cord syndrome should be investigated urgently.
- **MRI** (Fig. 28.47), for extent of compression & associated soft-tissue abnormality (Fig. 28.48).
- **Plain X-rays** for bony destruction & soft-tissue abnormalities.
- **Routine Ix**, including CXR, for evidence **systemic** disease.
- If **myelography** is performed, **CSF** should be taken for **analysis**.
- In cases of **complete spinal block**, this shows a normal cell count with a very elevated protein causing yellow discoloration of the fiuid (**Froin syndrome**).
- **The risk of acute deterioration after myelography** in spinal cord compression means that the neurosurgeons should be **alerted** before it is under taken.
- Where a secondary tumor is causing the compression, **needle biopsy** may be required to establish a tissue diagnosis.

Spinal Cord Compression

Symptoms	Signs	Investigation of acute spinal cord syndrome
Pain Localised over the spine or in a root distribution, which may be aggravated by coughing, sneezing or straining SC **Sensory** Paraesthesia, numbness or cold sensations, especially in the lower limbs, which spread proximally, often to a level on the trunk **Motor** Weakness, heaviness or stiffness of the limbs, most commonly the legs **Sphincters** Urgency or hesitancy of micturition, leading eventually to urinary retention	**Cervical, above C5** Upper motor neuron signs & sensory loss in all four limbs Diaphragm weakness (phrenic nerve) **Cervical, C5–T1** Lower motor neuron signs & segmental sensory loss in the arms; upper motor neuron signs in the legs Respiratory (intercostal) muscle weakness **Thoracic cord** Spastic paraplegia with a sensory level on the trunk Weakness of legs, sacral loss of sensation & extensor plantar responses **Cauda equina** Spinal cord ends approximately at the T12/L1 spinal level & spinal lesions below this level can cause lower motor neuron signs only by affecting the cauda equina	• **MRI of spine or myelography.** • **Plain X-rays of spine** • **Chest X-ray** • **Cerebrospinal Fluid** • **Serum vitamin B 12**

Spinal cord compression - Management

- Rx & prognosis depend on the nature of the underlying lesion.
- **Benign** tumors should be **excised**, & a good recovery is expected if treated early.
- **Extradural compression due to malignancy** is the most common cause of spinal cord compression in developed countries & has a **poor prognosis**.
- Useful function can be regained if Rx, such as **radiotherapy**, is initiated **within 24 h** of the onset of severe dificit; in collaborations with **oncologists** & **neurosurgeons**.
- **Tuberculosis** is common & may require **surgery** followed by appropriate **anti-TB**.
- **Traumatic lesions** require specialized neurosurgical Rx.

Conclusions

- The anatomy of the spine help a lot in lesion localization & shorten the way for investigations.
- Intrinsic & Extrinsic cord diseases each have may differential diagnosis.
- Stroke may cause ant. 2/3rd syndrome
- B12 deficiency is an important treatable cause
- Spinal cod compression must receive an important attention & treated urgently.

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Thannks